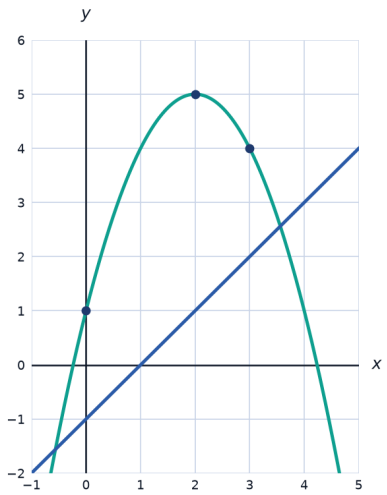
Exponential & Logarithmic Functions

ClipSAT Printable Practice Suite (clipsat.org) • Topic 5.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Multiplying and dividing rational expressions

For the model $f(x) = 5x + 6$ in multiplying and dividing rational expressions, evaluate $f(5)$.

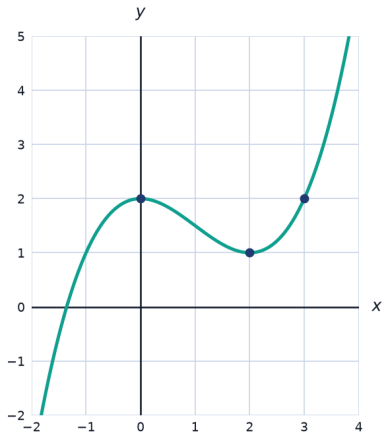


2 [Foundational] • Target: Multiplying and dividing rational expressions

Find the positive solutions to the quadratic equation $x^2 - 12x + 32 = 0$.

3 [Foundational] • Target: Solving equations with rational expressions

For the model $f(x) = 6x + 7$ in solving equations with rational expressions, evaluate $f(2)$.



4 [Foundational] • Target: Solving equations with rational expressions

Find the positive solutions to the quadratic equation $x^2 - 7x + 6 = 0$.

5 [Foundational] • Target: Solving equations with rational expressions

A data set recorded for solving equations with rational expressions consists of the four values $\{13, 17, 21, 27\}$. What is the arithmetic mean? Show your complete algebraic work.

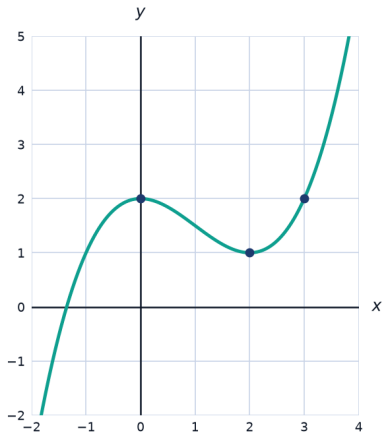
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6 [Foundational] • Target: Solving equations with rational expressions

For the model $f(x) = 6x + 7$ in solving equations with rational expressions, evaluate $f(3)$.

7 [Foundational] • Target: Solving equations with rational expressions

Find the positive solutions to the quadratic equation $x^2 - 9x + 14 = 0$.



Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Solving equations with rational expressions

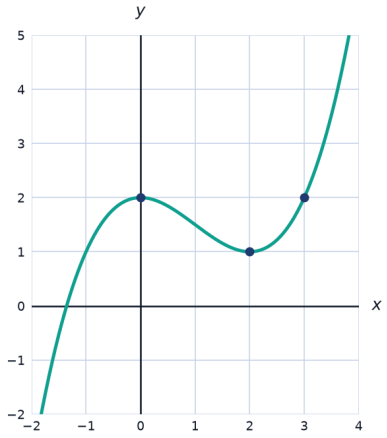
For the model $f(x) = 6x + 7$ in solving equations with rational expressions, evaluate $f(4)$.

9 [Intermediate] • Target: Solving equations with rational expressions

Find the positive solutions to the quadratic equation $x^2 - 11x + 24 = 0$.

10 [Intermediate] • Target: Solving equations with rational expressions

For the model $f(x) = 6x + 7$ in solving equations with rational expressions, evaluate $f(5)$.



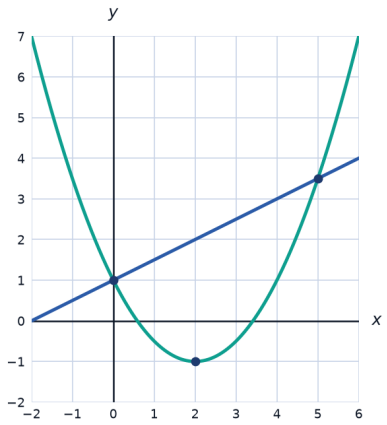
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11 [Intermediate] • Target: Solving equations with rational expressions

Find the positive solutions to the quadratic equation $x^2 - 13x + 36 = 0$.

12 [Intermediate] • Target: Graphical and analytical solutions to rational equations

For the model $f(x) = 2x + 8$ in graphical and analytical solutions to rational equations, evaluate $f(2)$.



13 [Intermediate] • Target: Graphical and analytical solutions to rational equations

A data set recorded for graphical and analytical solutions to rational equations consists of the four values $\{14, 18, 22, 28\}$. What is the arithmetic mean? Show your complete algebraic work.

14 [Intermediate] • Target: Graphical and analytical solutions to rational equations

For the model $f(x) = 2x + 8$ in graphical and analytical solutions to rational equations, evaluate $f(3)$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

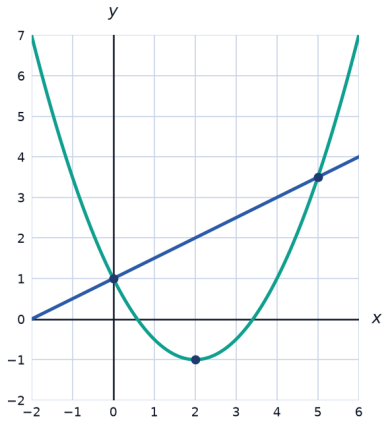
15 [Challenging] • Target: Graphical and analytical solutions to rational equations • Cross-Lesson: 1.01 Functions & Their Graphs

For the model $f(x) = 2x + 8$ in graphical and analytical solutions to rational equations, evaluate $f(4)$.

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16 [Challenging] • Target: Graphical and analytical solutions to rational equations • Cross-Lesson: 10.01 Matrices & Systems

For the model $f(x) = 2x + 8$ in graphical and analytical solutions to rational equations, evaluate $f(5)$.



17 [Challenging] • Target: Adding and subtracting rational expressions • Cross-Lesson: 11.01 Probability & Statistics

What is the sum of all real solutions to the equation $3^x + 3^{3-x} - 10 = |3^{x-1} - 3|$? Show your complete algebraic work.

18 [Challenging] • Target: Multiplying and dividing rational expressions • Cross-Lesson: 2.01 Quadratics & Complex Numbers

What is the sum of all real solutions to the equation $2^{\log_5(x^2)} - 6 \cdot x^{\log_5(2)} + 8 = 0$?.

19 [Challenging] • Target: Solving equations with rational expressions • Cross-Lesson: 3.01 Polynomial Functions

Find all solutions of $\frac{x^2+2x}{x^2-1} = 1 + \frac{3}{x-1}$.

20 [Challenging] • Target: Graphical and analytical solutions to rational equations • Cross-Lesson: 4.01 Rational Expressions

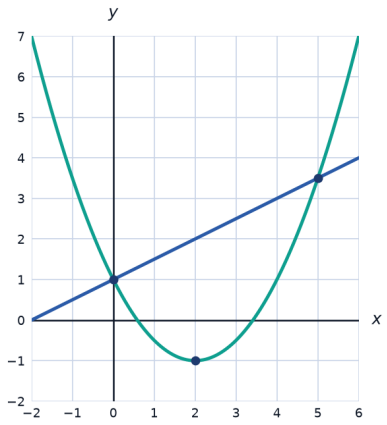
Find all solutions of $\ln x + \ln(x-2) = 0$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Simplifying rational expressions

For the model $f(x) = 3x + 4$ in simplifying rational expressions, evaluate $f(2)$.



a. 10

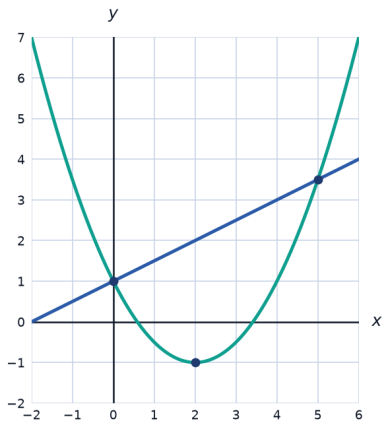
b. 13

c. 6

d. 14

22 [Foundational] • Target: Simplifying rational expressions

Find the positive solutions to the quadratic equation $x^2 - 6x + 8 = 0$.



a. $x = 2$ and $x = 5$

b. $x = 2$ and $x = 4$

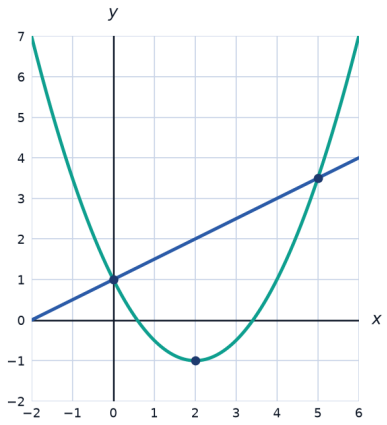
c. $x = -2$ and $x = -4$

d. $x = 6$ and $x = 8$

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23 [Foundational] • Target: Simplifying rational expressions

For the model $f(x) = 3x + 4$ in simplifying rational expressions, evaluate $f(5)$.



a. 19

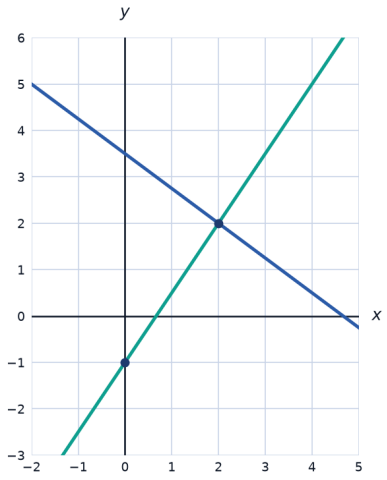
b. 22

c. 15

d. 23

24 [Foundational] • Target: Adding and subtracting rational expressions

For the model $f(x) = 4x + 5$ in adding and subtracting rational expressions, evaluate $f(2)$.



a. 13

b. 17

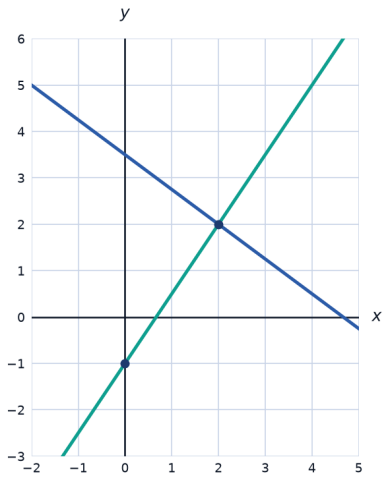
c. 8

d. 17

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25 [Foundational] • Target: Adding and subtracting rational expressions

Find the positive solutions to the quadratic equation $x^2 - 7x + 10 = 0$.



a. $x = 2$ and $x = 6$

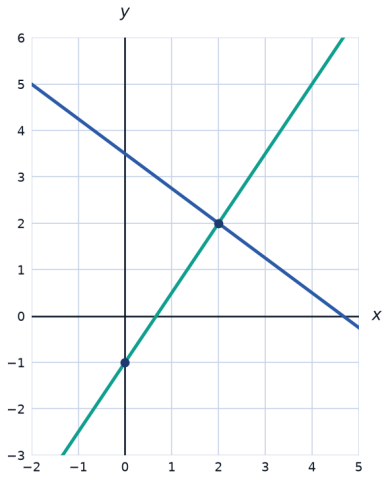
b. $x = 2$ and $x = 5$

c. $x = -2$ and $x = -5$

d. $x = 7$ and $x = 10$

26 [Foundational] • Target: Adding and subtracting rational expressions

For the model $f(x) = 4x + 5$ in adding and subtracting rational expressions, evaluate $f(5)$.



a. 25

b. 29

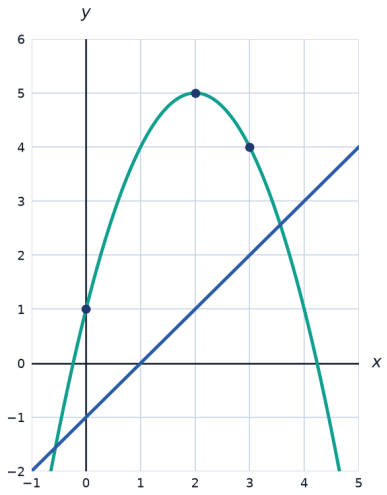
c. 20

d. 29

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27 [Foundational] • Target: Multiplying and dividing rational expressions

For the model $f(x) = 5x + 6$ in multiplying and dividing rational expressions, evaluate $f(2)$.



a. 16

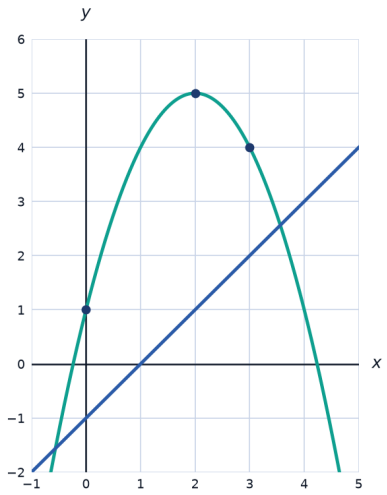
b. 21

c. 10

d. 20

28 [Foundational] • Target: Multiplying and dividing rational expressions

Find the positive solutions to the quadratic equation $x^2 - 8x + 12 = 0$.



a. $x = 2$ and $x = 7$

b. $x = 2$ and $x = 6$

c. $x = -2$ and $x = -6$

d. $x = 8$ and $x = 12$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Simplifying rational expressions

Find the positive solutions to the quadratic equation $x^2 - 4x + 3 = 0$.

a. $x = 1$ and $x = 4$

b. $x = 1$ and $x = 3$

c. $x = -1$ and $x = -3$

d. $x = 4$ and $x = 3$

30 [Intermediate] • Target: Simplifying rational expressions

A data set recorded for simplifying rational expressions consists of the four values $\{10, 14, 18, 24\}$. What is the arithmetic mean?

a. 14.5

b. 15.5

c. 16.5

d. 18.5

31 [Intermediate] • Target: Simplifying rational expressions

For the model $f(x) = 3x + 4$ in simplifying rational expressions, evaluate $f(3)$.

a. 13

b. 16

c. 9

d. 17

32 [Intermediate] • Target: Simplifying rational expressions

For the model $f(x) = 3x + 4$ in simplifying rational expressions, evaluate $f(4)$.

a. 16

b. 19

c. 12

d. 20

33 [Intermediate] • Target: Simplifying rational expressions

Find the positive solutions to the quadratic equation $x^2 - 8x + 15 = 0$.

a. $x = 3$ and $x = 6$

b. $x = 3$ and $x = 5$

c. $x = -3$ and $x = -5$

d. $x = 8$ and $x = 15$

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34 [Intermediate] • Target: Simplifying rational expressions

Find the positive solutions to the quadratic equation $x^2 - 10x + 24 = 0$.

a. $x = 4$ and $x = 7$

b. $x = 4$ and $x = 6$

c. $x = -4$ and $x = -6$

d. $x = 10$ and $x = 24$

35 [Intermediate] • Target: Adding and subtracting rational expressions

Find the positive solutions to the quadratic equation $x^2 - 5x + 4 = 0$.

a. $x = 1$ and $x = 5$

b. $x = 1$ and $x = 4$

c. $x = -1$ and $x = -4$

d. $x = 5$ and $x = 4$

36 [Intermediate] • Target: Adding and subtracting rational expressions

A data set recorded for adding and subtracting rational expressions consists of the four values $\{11, 15, 19, 25\}$. What is the arithmetic mean?

a. 15.5

b. 16.5

c. 17.5

d. 19.5

37 [Intermediate] • Target: Adding and subtracting rational expressions

For the model $f(x) = 4x + 5$ in adding and subtracting rational expressions, evaluate $f(3)$.

a. 17

b. 21

c. 12

d. 21

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Adding and subtracting rational expressions • Cross-Lesson: 2.01 Quadratics & Complex Numbers

Find the positive solutions to the quadratic equation $x^2 - 9x + 18 = 0$.

a. $x = 3$ and $x = 7$

b. $x = 3$ and $x = 6$

c. $x = -3$ and $x = -6$

d. $x = 9$ and $x = 18$

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39 [Challenging] • Target: Adding and subtracting rational expressions • Cross-Lesson: 3.01 Polynomial Functions

Find the positive solutions to the quadratic equation $x^2 - 11x + 28 = 0$.

a. $x = 4$ and $x = 8$

b. $x = 4$ and $x = 7$

c. $x = -4$ and $x = -7$

d. $x = 11$ and $x = 28$

40 [Challenging] • Target: Multiplying and dividing rational expressions • Cross-Lesson: 4.01 Rational Expressions

Find the positive solutions to the quadratic equation $x^2 - 6x + 5 = 0$.

a. $x = 1$ and $x = 6$

b. $x = 1$ and $x = 5$

c. $x = -1$ and $x = -5$

d. $x = 6$ and $x = 5$

41 [Challenging] • Target: Multiplying and dividing rational expressions • Cross-Lesson: 6.01 Radicals & Rational Exponents

A data set recorded for multiplying and dividing rational expressions consists of the four values $\{12, 16, 20, 26\}$. What is the arithmetic mean?

a. 16.5

b. 17.5

c. 18.5

d. 20.5

42 [Challenging] • Target: Multiplying and dividing rational expressions • Cross-Lesson: 7.01 Sequences & Series

For the model $f(x) = 5x + 6$ in multiplying and dividing rational expressions, evaluate $f(3)$.

a. 21

b. 26

c. 15

d. 25

43 [Challenging] • Target: Multiplying and dividing rational expressions • Cross-Lesson: 8.01 Conic Sections

For the model $f(x) = 5x + 6$ in multiplying and dividing rational expressions, evaluate $f(4)$.

a. 26

b. 31

c. 20

d. 30

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44 [Challenging] • Target: Multiplying and dividing rational expressions • Cross-Lesson: 9.01 Trigonometric Functions

Find the positive solutions to the quadratic equation $x^2 - 10x + 21 = 0$.

a. $x = 3$ and $x = 8$

b. $x = 3$ and $x = 7$

c. $x = -3$ and $x = -7$

d. $x = 10$ and $x = 21$

45 [Challenging] • Target: Solving equations with rational expressions • Cross-Lesson: 1.01 Functions & Their Graphs

A data set recorded for solving equations with rational expressions consists of the four values $\{13, 17, 21, 27\}$. What is the arithmetic mean?

a. 17.5

b. 18.5

c. 19.5

d. 21.5

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